Wang 2019 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells, gap-find) on one continuous 10x manifold; per-neighbourhood iLISI 1.03→2.60/7 across θ 2→10, plateaus (θ=20→2.59). θ=2 is a per-patient archipelago; θ=10 intermixes into one mass. 16 cells in 1 island removed → 9,634 cells, one component.

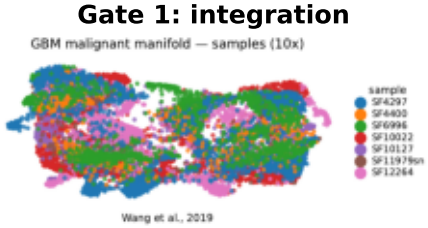
Gate 2 — module geography → GO (non-branching)

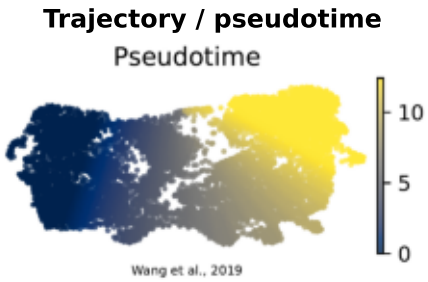
Shared Gast/pSom gradient (co-vary r=+0.16) with Crest an inverse terminal readout (Crest vs Gast r=−0.49, vs pSom r=−0.87). Crest-high decile at pseudotime 0.85 (terminal). No Gast/pSom competing arms; no Crest-on-Gast-arm inversion → no flag.

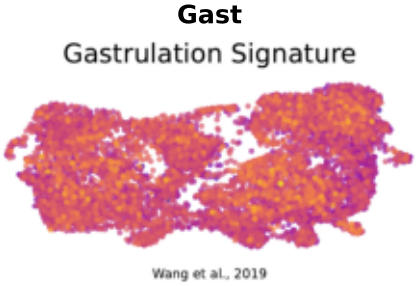
Gate 3 — hub cross-repression → NA

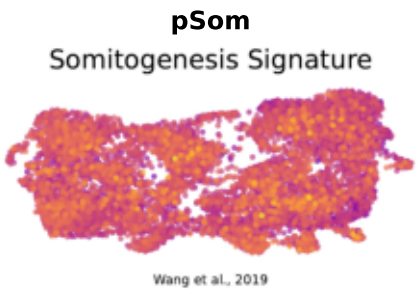
Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

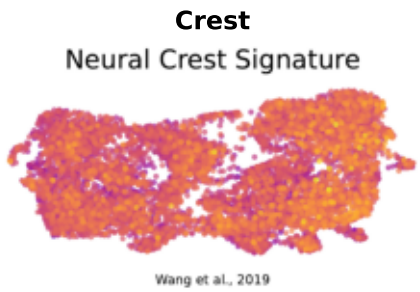
STEP FIGURES

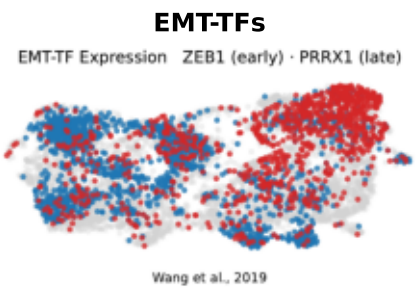












Parameter choices

Min cells / sample	≥237 (gap 396→143, ratio 2.77; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Overdispersed genes	2,113 (scFates GAM)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Island removal	6×-median radius graph; −16 → 9,634	**Root node**	Gast/pSom-high, Crest-low tip
Geometry	Linear (non-branching); fold-free 2.83	**EMT-TFs**	ZEB1 (early) → PRRX1 (late)

Dataset. Glioblastoma (Wang et al., 2019); 10x scRNA-seq, single leaf; 9,916 malignant raw → 9,634 final after threshold + island removal. IDH-wildtype, 7 samples.

Verdict. ✓ Pattern reproduces as a non-branching embryonic axis: a Gast/pSom-high origin grading to a Crest-high terminus (pSom ρ=−0.61, Crest ρ=+0.48 along pseudotime), continuity −0.98 checked on an independent diffusion embedding. Linear geometry — Gate 3 NA. EMT-TFs ZEB1 (early) → PRRX1 (late).

Gast μ = −0.003 pSom μ = +0.008 Crest μ = −0.007